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CMSC 204 UserAccessManager Project 3 Pseudocode

// pseudocode for GenericLinkedList.java

Declare a public class GenericLinkedList that implements Iterable<T>

Declare a private Node variable first
Declare a private Node variable last
Declare a private integer variable size initialized to 0

Declare a public constructor GenericLinkedList with no parameter arguments
Set size equal to 0
// end of constructor

Declare a private class Node

Declare a private T variable data
Declare a private Node variable next
Declare a private Node variable previous

Declare a private constructor Node with parameter argument data
Set this.data equal to data
Set this.next equal to null
Set this.previous equal to null
// end of Node constructor

// end of Node class

Declare a public T method addFirst with T parameter argument element

Declare Node newNode and set it equal to a new Node containing element

If isEmpty() is true
Set first equal to newNode
Set last equal to newNode
Else
Set newNode.next equal to first
Set first.previous equal to newNode
Set first equal to newNode
// end of if statement

Increment size by 1

Return element
// end of addFirst method

Declare a public T method addLast with T parameter argument element

Declare Node newNode and set it equal to a new Node containing element

If isEmpty() is true
Set first equal to newNode
Set last equal to newNode
Else
Set last.next equal to newNode
Set newNode.previous equal to last
Set last equal to newNode
// end of if statement

Increment size by 1

Return element
// end of addLast method

Declare a public T method remove with integer parameter argument index

If isEmpty() is true
Throw NoSuchElementException with message "The playlist is currently empty."

If index is less than 0 OR index is greater than or equal to size
Throw IndexOutOfBoundsException with message "Invalid index."

If index equals 0
Return removeFirst()

If index equals size minus 1
Return removeLast()

Declare Node currentNode equal to first

While i equals 0 to index - 1
Set currentNode equal to currentNode.next
Increment i by 1
// end of while/for loop

Declare Node previous equal to currentNode.previous
Declare Node next equal to currentNode.next

Set previous.next equal to next
Set next.previous equal to previous

Decrement size by 1

Return currentNode.data
// end of remove(index) method

Declare a public T method removeFirst with no parameter arguments

If isEmpty() is true
Throw NoSuchElementException with message "The playlist is currently empty."

Declare T data and set it equal to first.data

Set first equal to first.next

If first is not null
Set first.previous equal to null
Else
Set last equal to null

Decrement size by 1

Return data
// end of removeFirst method

Declare a public T method removeLast with no parameter arguments

If isEmpty() is true
Throw NoSuchElementException with message "The playlist is currently empty."

Declare T data and set it equal to last.data

Set last equal to last.previous

If last is not null
Set last.next equal to null
Else
Set first equal to null

Decrement size by 1

Return data
 // end of removeLast method

Declare a public boolean method remove with T parameter argument element

If isEmpty() is true
 Throw NoSuchElementException with message "The playlist is currently empty."

Declare Node currentNode equal to first

While currentNode is not null

If currentNode.data equals element

If currentNode equals first
 Call removeFirst()

Else if currentNode equals last
 Call removeLast()

Else
 Declare Node previous equal to currentNode.previous
 Declare Node next equal to currentNode.next

Set previous.next equal to next
 Set next.previous equal to previous

Decrement size by 1

Return true

Set currentNode equal to currentNode.next
 // end of while loop

Return false
 // end of remove(element) method

Declare a public boolean method contains with T parameter argument element

If isEmpty() is true
 Return false

Declare Node steppingNode equal to first

While steppingNode is not null

If steppingNode.data equals element
 Return true

Set steppingNode equal to steppingNode.next

// end of while loop

Return false
 // end of contains method

Declare a public T method get with integer parameter argument index

If isEmpty() is true
 Throw NoSuchElementException with message "The playlist is currently empty."

If index is less than 0 OR index is greater than or equal to size
 Throw IndexOutOfBoundsException with message "Invalid index."

Declare Node indexedNode equal to first

While i equals 0 to index - 1
 Set indexedNode equal to indexedNode.next

Increment i by 1
// end of while/for loop

Return indexedNode.data
// end of get method

Declare a public boolean method isEmpty with no parameter arguments

If size equals 0
Return true
Else
Return false
// end of isEmpty method

Declare a public T method getFirst with no parameter arguments

If isEmpty() is true
Throw NoSuchElementException with message "The playlist is currently empty."

Return first.data
// end of getFirst method

Declare a public T method getLast with no parameter arguments

If isEmpty() is true
Throw NoSuchElementException with message "The playlist is currently empty."

Return last.data
// end of getLast method

Declare a public void method clear with no parameter arguments

Set first equal to null
Set last equal to null
Set size equal to 0

// end of clear method

Declare a private class GenericIterator that implements ListIterator<T>

Declare Node currentNode equal to first
Declare Node lastReturn equal to null

Declare a public boolean method hasNext
If currentNode equals null
Return false
Else
Return true
// end of hasNext method

Declare a public T method next

If hasNext() equals false
Throw NoSuchElementException with message "There is no next element."

Set lastReturn equal to currentNode
Set currentNode equal to currentNode.next

Return lastReturn.data
// end of next method

Declare a public boolean method hasPrevious

If isEmpty() equals true
Return false

If currentNode equals null
If last is not null
Return true
Else
Return false

If currentNode.previous is not null
Return true
Else
Return false
// end of hasPrevious method

Declare a public T method previous

If hasPrevious() equals false
Throw NoSuchElementException with message "There is no previous element."

If currentNode equals null
Set currentNode equal to last
Else
Set currentNode equal to currentNode.previous

Set lastReturn equal to currentNode

Return lastReturn.data
// end of previous method

Declare a public void method remove

If lastReturn equals null
Throw IllegalStateException with message "There is nothing to remove."

Declare Node previous equal to lastReturn.previous
Declare Node next equal to lastReturn.next

If previous is not null
Set previous.next equal to next
Else
Set first equal to next

If next is not null
Set next.previous equal to previous
Else
Set last equal to previous

Decrement size by 1

Set lastReturn equal to null

// end of remove method

Declare a public integer method size

Return size

// end of size method

// end of GenericIterator class

Declare a public ListIterator<T> method iterator with no parameter arguments

Return a new GenericIterator object

// end of iterator method

Declare a public Object[] method toArray with no parameter arguments

Declare Object array linkedArray with size equal to size

Declare Node node equal to first

While i equals 0 to size - 1
 Set linkedArray[i] equal to node.data
 Set node equal to node.next
 Increment i by 1
 // end of for loop

Return linkedArray

// end of toArray method

// end of class GenericLinkedList and GenericLinkedList.java pseudocode

// pseudocode for Song.java

Declare a public class Song

Declare a private String variable songName
Declare a private String variable artistName

Declare a public constructor Song with no parameter arguments

 Set songName equal to null
 Set artistName equal to null

// end of constructor

Declare a public constructor Song with String parameter argument songName and String parameter argument artistName

Set this.songName equal to songName
Set this.artistName equal to artistName

// end of constructor

Declare a public void method setSongName with String parameter argument songName

Set this.songName equal to songName

// end of setSongName method

Declare a public String method getSongName with no parameter arguments

Return songName

// end of getSongName method

Declare a public void method setArtistName with String parameter argument artistName

Set this.artistName equal to artistName

// end of setArtistName method

Declare a public String method getArtist with no parameter arguments

Return artistName

// end of getArtist method

Declare a public boolean method equals with Object parameter argument object

If this object is equal to object
Return true

If object is not an instance of Song
Return false

Declare Song variable object2 and set it equal to object cast as Song

If songName equals object2.songName AND artistName equals object2.artistName
Return true
Else
Return false

// end of equals method

Declare a public String method toString with no parameter arguments

Return songName + " - " + artistName

// end of toString method

Declare a public Object method getTitle with no parameter arguments

Return songName

// end of getTitle method

// end of class Song and Song.java pseudocode

// pseudocode for User.java

Declare a public class User

Declare a private integer variable playlistCount initialized to 0
Declare a private String variable username
Declare a private String variable password

Declare a private GenericLinkedList<Playlist> variable userPlaylists

Declare a public constructor User with no parameter arguments

Set playlistCount equal to 0
Set username equal to null
Set password equal to null
Set userPlaylists equal to a new GenericLinkedList object

// end of constructor

Declare a public constructor User with String parameter arguments username and password

Set this.username equal to username
Set this.password equal to password
Set userPlaylists equal to a new GenericLinkedList object

// end of constructor

Declare a public constructor User with integer parameter argument playlistCount, String parameter argument username, and String parameter argument password

Set this.playlistCount equal to playlistCount
Set this.username equal to username
Set this.password equal to password
Set userPlaylists equal to a new GenericLinkedList object

// end of constructor

Declare a public void method addPlaylist with Playlist parameter argument playlist

Call method addLast Playlist with argument playlist

Increment playlistCount by 1

// end of addPlaylist method

Declare a public integer method getPlaylistCount with no parameter arguments

Return playlistCount

// end of getPlaylistCount method

Declare a public void method removePlaylist with Playlist parameter argument playlist

Decrement playlistCount by 1

// end of removePlaylist method

Declare a public boolean method emptyCollection with no parameter arguments

If playlistCount equals 0
Return true
Else
Return false

// end of emptyCollection method

Declare a public String method getUsername with no parameter arguments

Return username

// end of getUsername method

Declare a public String method getPassword with no parameter arguments

Return password

// end of getPassword method

Declare a public GenericLinkedList<Playlist> method getPlaylists with no parameter arguments

Return userPlaylists

// end of getPlaylists method

Declare a public void method setPassword with String parameter argument password2

Set this.password equal to password2

// end of setPassword method

// end of class User and User.java pseudocode

// pseudocode for Playlist.java

Declare a public class Playlist

Declare a private String variable name
Declare a private GenericLinkedList<Song> variable userSongs
Declare a private ListIterator<Song> variable userIterator
Declare a private Song variable currentSong

Declare a public constructor Playlist with no parameter arguments

Set name equal to null
Set userSongs equal to a new GenericLinkedList object
Set userIterator equal to iterator from userSongs
Set currentSong equal to null

// end of constructor

Declare a public constructor Playlist with String parameter argument name

Set this.name equal to name
Set userSongs equal to a new GenericLinkedList object
Set userIterator equal to iterator from userSongs
Set currentSong equal to null

// end of constructor

Declare a public boolean method addSong with Song parameter argument song

Call addLast on userSongs with song as the argument

If currentSong equals null
Set currentSong equal to song

Reset userIterator by assigning it to userSongs.iterator()

Return true

// end of addSong method

Declare a public Song method getCurrentSong with no parameter arguments

If isEmpty() equals true
Throw NoSuchElementException with message "The playlist is currently empty."

Return currentSong

// end of getCurrentSong method

Declare a public integer method getSize with no parameter arguments

Declare integer variable size
Set size equal to userSongs.size()

Return size

// end of getSize method

Declare a public boolean method isEmpty with no parameter arguments

If userSongs.size() equals 0
Return true
Else
Return false

// end of isEmpty method

Declare a public GenericLinkedList<Song> method getSongs with no parameter arguments

Return userSongs

// end of getSongs method

Declare a public String method getName with no parameter arguments

Return name

// end of getName method

Declare a public Song method nextSong with no parameter arguments

If isEmpty() equals true
Throw NoSuchElementException with message "The playlist is currently empty."

If userIterator.hasNext() equals true
Set currentSong equal to userIterator.next()
Return currentSong
Else
Return null

// end of nextSong method

Declare a public Song method previousSong with no parameter arguments

If isEmpty() equals true
Throw NoSuchElementException with message "The playlist is currently empty."

If userIterator.hasPrevious() equals true
Set currentSong equal to userIterator.previous()
Return currentSong
Else
Return null

// end of previousSong method

Declare a public boolean method removeSong with Song parameter argument song

If isEmpty() equals true
Throw NoSuchElementException with message "The playlist is currently empty."

If userSongs.remove(song) equals true
Return true

Else
Return false

// end of removeSong method

// end of class Playlist and Playlist.java pseudocode

// pseudocode for SpotifyManager.java

Declare a public class SpotifyManager

Declare a private GenericLinkedList<User> variable usersList initialized to a new GenericLinkedList object

Declare a public void method loadUsersFromFile with String parameter argument filename
Throws IOException and InvalidUserFormatException

Declare BufferedReader variable userReader and set it equal to a new BufferedReader using FileReader with filename

Declare String variable line
Declare User variable currentUser initialized to null
Declare Playlist variable currentPlaylist initialized to null

While reading each line from userReader is not null

Set line equal to trimmed version of line

If line equals "# USER"
Set currentUser equal to null
Set currentPlaylist equal to null

Else if line starts with "username:"

Declare String variable username
Set username equal to substring of line starting at index 9 and trim whitespace

Set currentUser equal to a new User object with username and null password

Else if line starts with "password:"

Declare String variable password
Set password equal to substring of line starting at index 9 and trim whitespace

If currentUser equals null
Throw InvalidUserFormatException with message "User format is invalid."

Call setPassword on currentUser with password

Add currentUser to usersList using addLast

Else if line starts with "playlist:"

Declare String variable playlistName
Set playlistName equal to substring of line starting at index 9 and trim whitespace

Set currentPlaylist equal to a new Playlist object with playlistName

If currentUser is not null
Call addPlaylist on currentUser with currentPlaylist

Else if line starts with "song:"

Declare String variable songData
Set songData equal to substring of line starting at index 5 and trim whitespace

Declare String array songParts and set it equal to splitting songData using "-"

If length of songParts is not equal to 2
Throw InvalidUserFormatException with message "User format is invalid."

Declare String variable songTitle equal to trimmed songParts[0]
Declare String variable songArtist equal to trimmed songParts[1]

Declare Song variable song and set it equal to a new Song object using songTitle and songArtist

If currentPlaylist is not null
Call addSong on currentPlaylist with song

// end of while loop

Close userReader

// end of loadUsersFromFile method

Declare a public User method findUser with String parameter arguments username and password
Throws UserNotFoundException and InvalidUserFormatException

Declare ListIterator<User> variable userIterator equal to usersList.iterator()

While userIterator.hasNext() equals true

Declare User variable user equal to userIterator.next()

If user.getUsername() equals username

If user.getPassword() does not equal password
Throw InvalidPasswordException with message "Password is invalid."

Return user

// end of while loop

Throw UserNotFoundException with message "User cannot be found."

// end of findUser method

Declare a public User[] method getUsers with no parameter arguments

Declare integer variable size equal to usersList.size()

Declare User array userArray with size equal to size

Declare ListIterator<User> variable userIterator equal to usersList.iterator()

Declare integer variable i initialized to 0

While userIterator.hasNext() equals true

Set userArray[i] equal to userIterator.next()

Increment i by 1

// end of while/for loop

Return userArray

// end of getUsers method

// end of class SpotifyManager and SpotifyManager.java pseudocode

// pseudocode for UserNotFoundException.java

Declare a public class UserNotFoundException that extends Exception

Declare a public constructor UserNotFoundException with String parameter message

Call the superclass constructor Exception and pass message as the argument

// end constructor

// end class UserNotFoundException

// pseudocode for InvalidPasswordException.java

Declare a public class InvalidPasswordException that extends RuntimeException

Declare a public constructor InvalidPasswordException with String parameter message

Call the superclass constructor RuntimeException and pass message as the argument

// end constructor

// end class InvalidPasswordException

// pseudocode for InvalidUserFormatException.java

Declare a public class InvalidUserFormatException that extends RuntimeException

Declare a public constructor InvalidUserFormatException with String parameter message

Call the superclass constructor RuntimeException and pass message as the argument

// end constructor

// end class InvalidUserFormatException